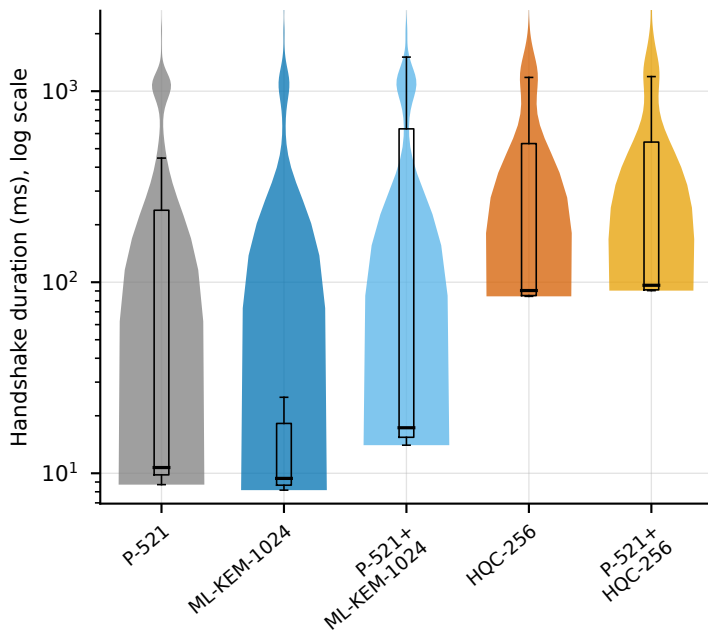


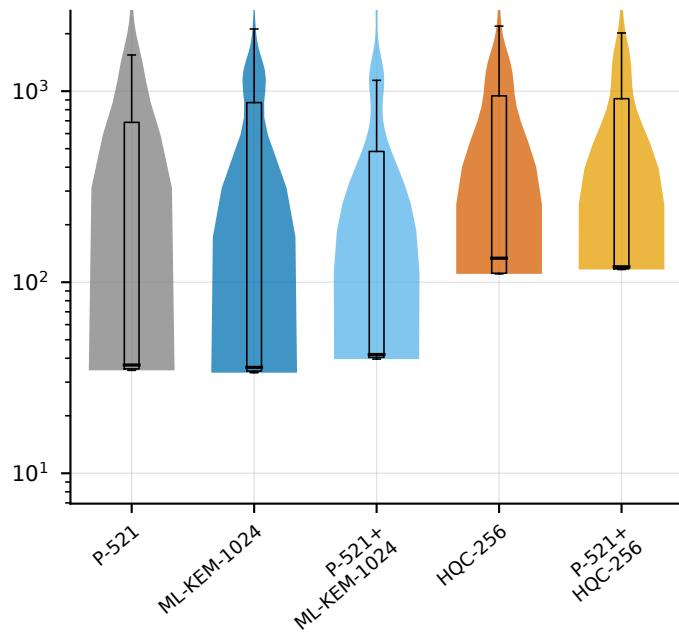
(a)

secp521r1 (classical)



(b)

mldsa87 (post-quantum)



(c)

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	288.2	184.8	5.32k	534.1	281.5	27.49k
ML-KEM-1024	251.2	255.2	6.44k	477.5	197.7	13.82k
P-521+ML-KEM-1024	352.1	184.9	7.04k	437.7	179.3	7.24k
HQC-256	506.7	165.3	9.64k	670.5	170.1	14.46k
P-521+HQC-256	539.4	164.9	7.77k	676.4	169.2	13.74k